Matthew Dong

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Experience

NJStat — **Bridgewater, NJ**
Artificial Intelligence Intern — Jan 2025 – Present

- Updated internal AI software using the **Transformers** library in **Python** to improve employee monitoring by 20%.
- Adapted to new situations resulting from doing development directly in a Windows environment instead of Linux.

Colgate-Palmolive — **Piscataway, NJ**
Application and Software Development Intern (E-Commerce DevOps) — Jun 2022 – Dec 2024

- Developed **Python** scripts that utilize **cloud functions** and **API gateways** to parse **JSON** alerts for various applications and send them to Google Chat spaces using **webhooks**, clearing cluttered email inboxes.
- Refactored code to fit company standards and allowed non-technical staff to easily modify parameters in the future.
- Completed new workflows and modified existing ones with a focus on reusability, scalability, and efficiency.
- Experienced different aspects of company culture and worked in a non-school environment by participating in meetings and code reviews, as well as making presentations to both technical and non-technical staff.

Sanofi — **Bridgewater, NJ**
Data Analyst Intern — Dec 2020 – Feb 2021

- Utilized **R** in order to analyze COVID-19 data, while keeping statistical principles in mind.
- Modified and analyzed movie data utilizing machine learning topics: **PCA**, **bootstrapping**, and **decision trees.**

Personal Projects

(More projects available on GitHub: github.com/Matt-J-Dong)

VentureAI (Python: PyTorch)

- Developed an automated travel agent that utilizes fine-tuned **open-source LLMs** that provides personalized travel recommendations tailored to each user's preferences.
- Addressed limitations of state of the art models such as hallucination and implemented key technical innovations such as **LoRA** and **quantization**.

Multicore Chess (C++: OpenMP)

- Greatly improved the speed of move calculations, achieving up to **12x** speedup in comparison to the sequential version.
- Optimized the evaluation function and **parallel alpha-beta pruning** algorithms despite limited CPUs and no access to GPUs.

Top-Towns-To-Take-Over-Tech (Spark Scala, Java MapReduce, Shell)

- Built an analytic with a group to find out which American cities are the best for tech careers.
- We cleaned, profiled, and performed ETL on our data before joining all the datasets together, and displayed our results through a Tableau visualization and a Hive warehouse.

Chinook Database (SQL, SQLite)

- Used SQL to query the chinook database for specific necessary information. Key aspects include joins, using groups to get averages, and sorting.

Education

New York University — **New York, NY**
M.S. Computer Science — GPA: 3.95/4.00

- Coursework: DevOps and Agile Methodologies, Big Data Application Development, Database Systems, Programming Languages, Multicore Processors, Deep Learning

New York University — **New York, NY**
B.A. Computer Science; B.A. Data Science — GPA: 3.93/4.00

- Coursework: Fundamentals of Machine Learning, Advanced Topics: Deep Learning, Responsible Data Science, Basic Algorithms, Operating Systems, Big Data for Analytics Applications, Linear Algebra, Prob and Stat

Skills

Languages: Python, R, Markdown, C++, Scala, SQL, R, Bash, Java, x86-64 Assembly
Libraries: PyTorch, Transformers, Wandb, NumPy, Pandas, Matplotlib, Seaborn, Scikit-Learn, OpenMP, MPI
Technologies: GitHub, GitHub Actions, Hadoop, K8s, Docker, Google Cloud Platform, Jira, Confluence, Regex

Organizations/Leadership

Intern Social Committee — **Piscataway, NJ**
Head of Internal Operations — Jun 2022 – Dec 2024

- Wrote meeting agendas, coordinated and led weekly meetings, and delegated tasks to achieve the end goal of the committee was to design fun programs for the interns of Colgate to enjoy, received highly positive feedback from the attendees of events.